CR-015: Client-Side Search and Filtering of the Admin Student List

1. Change Request Details

Field	Details
Change Request ID	CR-015
Title	Student Records Export to CSV
Priority	Medium
Difficulty	Medium
Change Type	Full-Stack Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing student records and student management functionality

2. Business Requirement

Administrators may need to download and use student records outside the Student Management System for administrative, reporting, or record-keeping purposes.

Currently, student information is available within the system, but there is no dedicated facility for administrators to export the student records into a portable file format.

The system should provide an easy and reliable way for authorized administrators to export student records as a CSV file.

3. Current Behaviour

- The administrator can view student records within the Student Management System.
- Student records are displayed within the existing student management interface.
- No CSV export facility is currently available.
- Administrators must manually view or copy student information if they need an external record.
- Existing student management and CRUD operations use the existing student identifiers.
- Existing administrator authentication and authorization are already used to protect student management functionality.

4. Requested Change

Add a **Student Records Export** facility to the administrator's student management interface.

The implementation must:

- Provide an **Export CSV** button for authorized administrators.
- Export the available student records into CSV format.

- Include relevant student information such as:
    - Student ID
    - Registration Number
    - First Name
    - Last Name
    - Email
    - Phone Number
- Generate a properly formatted CSV file.
- Download the generated file to the administrator's device.
- Use a meaningful filename such as students.csv.
- Include CSV column headers.
- Handle multiple student records correctly.
- Handle an empty student list without causing an application error.
- Preserve existing student viewing, editing, and deleting functionality.
- Maintain existing administrator authentication and authorization.
- Ensure the export functionality does not expose student records to unauthorized users.

5. Scope

In Scope

- Export CSV button in the administrator student management interface.
- Retrieval of existing student records for export.
- CSV file generation.
- CSV column headers.
- Export of relevant student information.
- CSV file download.
- Meaningful CSV filename.
- Empty-record handling.
- Administrator authorization.
- Preservation of existing CRUD functionality.
- Basic responsive presentation of the export control.

Out of Scope

- PDF export.
- Excel/XLSX export.
- Emailing exported files.
- Cloud storage integration.
- Scheduled automatic exports.
- Changes to the existing student database structure.
- Changes to authentication or authorization rules.
- Changes to existing student CRUD operations.

6. Expected / Potentially Affected Files

The exact files should be confirmed during implementation and impact analysis.

Potentially affected files include:

- frontend/src/pages/DashboardPage.jsx
- Existing dashboard/student management stylesheet
- Backend student controller
- Backend student routes
- Backend student model, if required for export data retrieval

Students may modify additional files if required. Any additional files must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

- FR-001: An Export CSV control shall be available to authorized administrators.
- FR-002: The system shall export the available student records in CSV format.
- FR-003: The exported CSV shall contain appropriate column headers.
- FR-004: The CSV shall include Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
- FR-005: Multiple student records shall be exported correctly.
- FR-006: The downloaded file shall use a meaningful filename.
- FR-007: The exported data shall accurately reflect the student records available in the system.
- FR-008: An empty student list shall be handled without causing a runtime error.
- FR-009: Existing student viewing, editing, and deleting functionality shall continue to work.
- SEC-001: Only authenticated and authorized administrators shall be able to access the student export functionality.
- SEC-002: The export feature shall not bypass existing authentication or authorization controls.
- SEC-003: The feature shall not expose student records to unauthorized users.
- SEC-004: Existing administrator security controls shall remain unchanged.

8. Acceptance Criteria

The CR is successfully implemented when:

- AC-001: An authorized administrator can see the Export CSV option.
- AC-002: An administrator can successfully download a CSV file containing student records.
- AC-003: The exported CSV contains the required column headers.
- AC-004: The CSV contains the correct Student ID, Registration Number, First Name, Last Name, Email, and Phone Number values.
- AC-005: Multiple student records are exported correctly.
- AC-006: The downloaded file has the expected .csv file extension and meaningful filename.
- AC-007: Exported data matches the student records displayed/available in the system.
- AC-008: An empty student list is handled without a runtime error.
- AC-009: Unauthorized users cannot access the student export functionality.
- AC-010: Existing student CRUD functionality continues to work correctly.
- AC-011: Existing administrator authentication and authorization continue to work correctly.
- AC-012: Exporting student records does not modify or delete the original student records.

9. Expected Outcome

After implementation:

- Administrators can export student records from the Student Management System.
- The exported records are provided in CSV format.
- The CSV contains the required student information and appropriate headers.
- The downloaded file can be opened using spreadsheet software.
- Exported information accurately represents the available student records.
- Empty student data is handled safely.
- Unauthorized users cannot access the export functionality.
- Existing student CRUD functionality remains unaffected.
- Existing authentication and authorization remain intact.
- Exporting data does not modify the original student records.

10. Implementation Report Requirement

After implementation, the Final Exam Software Change Implementation Report shall include:

- Implementation summary.
- Business requirement and purpose of the change.
- Affected files.
- Impact analysis.
- Important technical and security changes.
- CSV export implementation details.
- Test cases and results.
- Acceptance criteria verification.
- Relevant screenshots/evidence, including:
    - Student management page with **Export CSV** button.
    - Successful CSV download.
    - Opened CSV file showing exported student records.
    - Authorization/security testing where applicable.
- Final status and outstanding issues.